DraftBench — Special Education Resource

Curriculum Aligned & Autism Support Framework

Kinematics Basics

Subject: Physics | **Grade:** 11 | **Language:** ENGLISH | **Duration:** 30 mins

CDC Outcome: Physics 11: Understand the basic concepts of kinematics

Visual Schedule

• **First** (5 mins): clock

• **Then** (10 mins): book

• **Last** (10 mins): checkmark

• **Done** (5 mins): smiley face

Lesson Slides



Slide 1: Introduction to Kinematics

• Understand motion basics

• Objects move in space

• Study without forces

Kinematics is about how objects move in space and time.

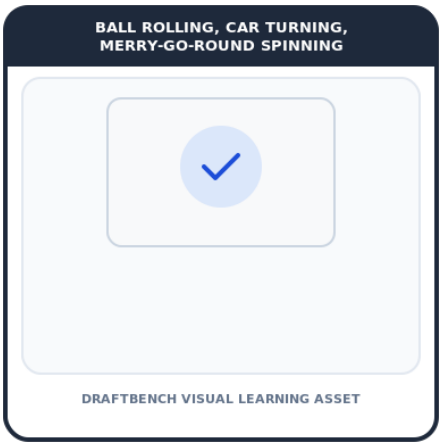
Example: Car moving on a road

Key words: motion: object's change in position

Teacher Script: "Today we learn about how objects move."

Communication Access: point to pictures

Sensory Note: null



Slide 2: Types of Motion

• Straight line motion

• Curved line motion

• Circular motion

Objects can move in straight, curved, or circular paths.

Example: Ball rolling on floor, car turning, merry-go-round

Key words: straight line: linear motion

Teacher Script: "Objects can move in different ways."

Communication Access: point to pictures

Sensory Note: null



Slide 3: Distance and Displacement

• Distance is total path

• Displacement is shortest path

Distance is the total path length, displacement is the shortest path.

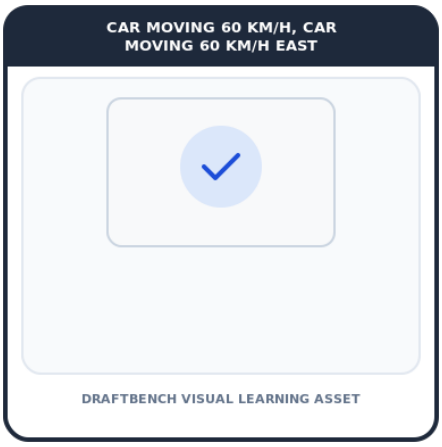
Example: Walking around a square, walking directly to corner

Key words: distance: total path length; displacement: shortest path length

Teacher Script: "Distance and displacement are different."

Communication Access: point to pictures

Sensory Note: null



Slide 4: Speed and Velocity

• Speed is how fast

• Velocity includes direction

Speed is how fast an object moves, velocity includes direction.

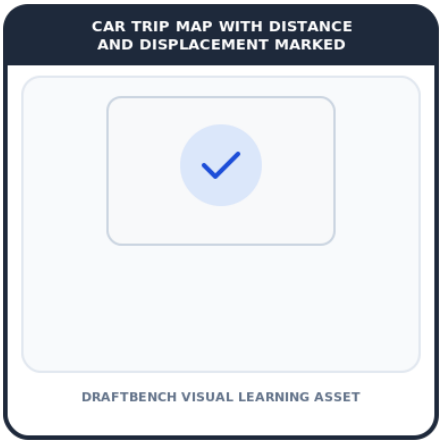
Example: Car moving 60 km/h, car moving 60 km/h east

Key words: speed: rate of change of distance; velocity: speed with direction

Teacher Script: "Speed and velocity are different."

Communication Access: point to pictures

Sensory Note: null



Slide 5: Practical Example

• Calculate distance

• Find displacement

• Determine speed

Apply kinematics concepts to real-life situations.

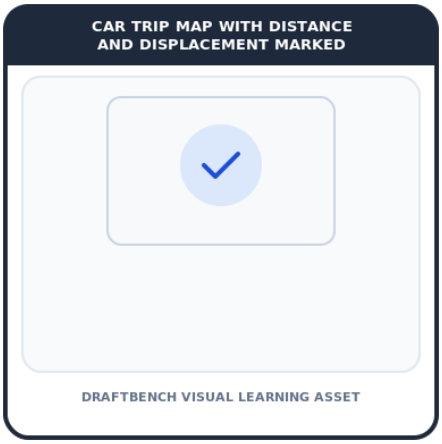
Example: Calculate distance, displacement, speed for a car trip

Key words: distance: total path length; displacement: shortest path length; speed: rate of change of distance

Teacher Script: "Let's solve a practical example."

Communication Access: point to pictures

Sensory Note: null



Slide 6: Review Kinematics

• Types of motion

• Distance and displacement

• Speed and velocity

Review the key concepts of kinematics.

Example: Discuss car trip example

Key words: motion: object's change in position; distance: total path length; displacement: shortest path length; speed: rate of change of distance; velocity: speed with direction

Teacher Script: "Let's review what we learned today."

Communication Access: point to pictures

Sensory Note: null

IEP Target

Goal 1

• **English Goal:** Student will answer 5 given topics correctly 80% of the time without errors

• **Nepali Goal:** स्टूडेंट नेपाली में 80% गलती न करके दिए गए 5 वषियों पर सही जवाब दे सकें

• **Baseline Probe:** Student currently answers 60% of the given topics correctly

• **Measurable Criteria:** 80% accuracy across 5 consecutive sessions

• **Progress Monitoring:** Collect data after every session

Accommodations

• Use of visual aids and diagrams

Modifications

• Reduce the number of questions per session

Prompt Hierarchy

independent verbal visual gestural physical

Reinforcement Plan

• **Preferred Reinforcer:** Playing a game

• **Schedule:** token board every 2 correct responses

• **First–Then:** First complete the task, then play a game

Sensory Accommodations

• Use of calming music in the background

Calm-Down Strategy

Take deep breaths and count to 10

Local Context Used

• Use of car trips as examples

Curriculum Sources

No sources retrieved.